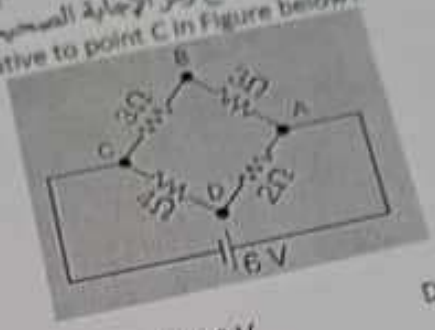
Physical Constants

Coulomb constant $k = \frac{1}{4\pi\epsilon_0} = 8.987 \times 10^9\ N.\frac{m^2}{C^2}$ Permittivity of free space $\epsilon_0 = 8.854 \times 10^{-12}\ C^2/N.m^2$.
Electron mass $= 9.109 \times 10^{-31}\ kg$. Proton mass $= 1.672 \times 10^{-27}\ kg$.
Electron charge $= 1.602 \times 10^{-19}\ C$ The mathematical constant $\pi = 3.14$
Acceleration due to gravity $g = 9.80\ m/s^2$

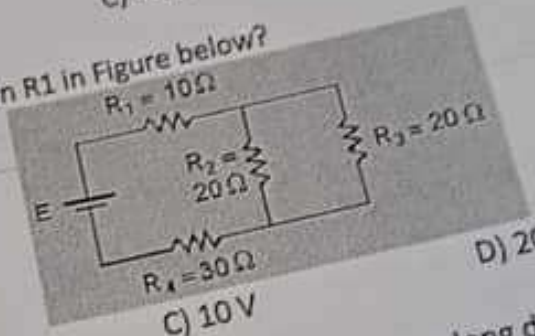
الجزء الأول يتكون من (16) سؤال إختيار من متعدد ووضع رمز الإجابة الصحيحة في الجدول الموجود في صفحة الغلاف

Q1) What is the potential of point B relative to point C in Figure below?



A) +6.0 V B) +4.0 V C) +3.0 V D) +2.0 V

Q.2) If E = 40 V, what is the voltage on R1 in Figure below?



A) 6.7 V B) 8.0 V C) 10 V D) 20 V

Q.3) A 2.0 μF capacitor is charged through a 50 kΩ resistor. How long does it take for the capacit 90% of full charge?

A) 0.923 s B) 0.23 s C) 0.16 s D) 2.92 s

Q.4) An electron follows a circular path (radius = 15 cm) in a uniform magnetic field of magn $(3 \times 10^{-4}\ T)$. What is the period of this motion?

A) 0.12 μs B) 1.6 ms C) 0.33 μs D) 3.8 ms

Q.5) A particle ($q = -4.0\ \mu C$, $m = 5.0$ mg) moves in a uniform magnetic field with a velo of 2.0 km/s and a direction that is 50° away from that of the magnetic field. The partic acceleration with a magnitude of 5.8 m/s². What is the magnitude of the magnetic fi

A) 5.3 T B) 3.6 T C) 7.1 T D) 4.7 T

Q.6) A light bulb is rated at 30 W when operated at 120 V. How much charge ente in 1.0 min?

A) 17 C B) 15 C C) 14 C D) 13

Q.9) By what factor is the capacitance of a metal sphere multiplied if its volume is tripled?

A) 3 B) $(3)^{-\frac{1}{3}}$ C) $(3)^{\frac{1}{3}}$ D) $\frac{1}{3}$

Q.10) When a battery is connected to the plates of a 3.00 μF capacitor, it stores a charge of 27.0 μC. What is the voltage of the battery?

A) 3 V B) 12 V C) 6 V D) 9 V

Q.11) An air-filled spherical capacitor is constructed with inner- and outer-shell radii of 7.00 cm and 14.0 cm, respectively. Calculate the capacitance of the device.

A) 15.6 pF B) 12.2 pF C) 6.3 pF D) 9.7 pF

Q.12) The electric potential at x = 3.00 m is 120 V, and the electric potential at x = 5.00 m is 190 V. What is the x component of the electric field in this region, assuming the field is uniform?

A) +17.5 N/C B) -17.5 N/C C) +35.0 N/C D) -35.0 N/C

13) A proton is released from rest at the origin in a uniform electric field in the positive x direction with magnitude 850 N/C. What is the change in the electric potential energy of the proton–field system when the proton travels to x = 2.50 m?

A) -3.4×10^{-16} J B) $+3.4 \times 10^{-16}$ J C) -2.2×10^{-16} J D) $+2.2 \times 10^{-16}$ J

Object A has a charge of +2 μC, and object B has a charge of +6 μC. Which statement is true about the electric forces on the objects?

A) $F_{AB} = -F_{BA}$ B) $F_{AB} = 0.5F_{BA}$ C) $F_{AB} = 3F_{BA}$ D) $3F_{AB} = F_{BA}$

charge of 170 μC is at the center of a cube of edge 80 cm. No other charges are nearby. Find the flux (in N·m²/C) through the whole surface of the cube.

A) $\times 10^7$ B) 2.48×10^7 C) 1.92×10^7 D) 7.2×10^7

2.0 V battery is connected to a capacitor, resulting in 54.0 μC of charge stored on the capacitor. How much energy is stored in the capacitor?

A) 10^{-4} J B) 3.24×10^{-4} J C) 12.2×10^{-4} J D) 2×10^{-4}

Q.7) A rod of 2 m length and a square (2.0 mm × 2.0 mm) cross section is made of a material with a resistivity of $(6 \times 10^{-8}\ \mathbf{Ohm \cdot m})$. If a potential difference of 0.50 V is placed across the ends of the rod, what power generated in the rod?

A) 3.0 W B) 5.3 W C) 8.3 W D) 16.3 W

Q.8) The radius of 22-gauge Nichrome wire is 0.32 mm. If a potential difference of 10 V is maintained across a 1.0 m length of the Nichrome wire, what is the current in the wire?

$(\rho_{Nichrome} = 1 \times 10^{-6}\ Ohm \cdot meter)$

A) 3.2 A B) 7.3 A C) 6.4 A D) 1.34 A

Q.9) By what factor is the capacitance of a metal sphere multiplied if its volume is tripled?

A) 3 B) $(3)^{-\frac{1}{3}}$ C) $(3)^{\frac{1}{3}}$ D) $\frac{1}{3}$

Q.10) When a battery is connected to the plates of a 3.00 μF capacitor, it stores a charge of 27.0 μC. What is the voltage of the battery?

A) 3 V B) 12 V C) 6 V D) 9 V

Q.11) An air-filled spherical capacitor is constructed with inner- and outer-shell radii of 7.00 cm and 14.0 cm, respectively. Calculate the capacitance of the device.

A) 15.6 pF B) 12.2 pF C) 6.3 pF D) 9.7 pF

Q.12) The electric potential at x = 3.00 m is 120 V, and the electric potential at x = 5.00 m is 190 V. What is the x component of the electric field in this region, assuming the field is uniform?

A) +17.5 N/C B) -17.5 N/C C) +35.0 N/C D) -35.0 N/C

Q.13) A proton is released from rest at the origin in a uniform electric field in the positive x direction of magnitude 850 N/C. What is the change in the electric potential energy of the proton–field system when the proton travels to x = 2.50 m?

A) -3.4×10^{-16} J B) $+3.4 \times 10^{-16}$ J C) -2.2×10^{-16} J D) $+2.2 \times 10^{-16}$ J

Q.14) Object A has a charge of +2 μC, and object B has a charge of +6 μC. Which statement is true about the electric forces on the objects?

A) $F_{AB} = -F_{BA}$ B) $F_{AB} = 0.5F_{BA}$ C) $F_{AB} = 3F_{BA}$ D) $3F_{AB}$

Q.15) A charge of 170 μC is at the center of a cube of edge 80 cm. No other charges are nearby. Find the net electric flux $(\mathrm{N \cdot m^2/C})$ through the whole surface of the cube.

A) 5.2×10^{7} B) 2.48×10^{7} C) 1.92×10^{7} D) 7.

Q.16) A 12.0 V battery is connected to a capacitor, resulting in 54.0 μC of charge stored on the capacitor. How much energy is stored in the capacitor?

A) 5.41×10^{-4} J B) 3.24×10^{-4} J C) 12.2×10^{-4} J D) 4.3